

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Student Profile Generator Using ES6 Classes and DOM Manipulation

2. AIM

To build an interactive Student Profile Generator web application using ES6 JavaScript class syntax, event handling, and dynamic DOM element creation.

3. OBJECTIVE

The objectives of this experiment are:

- Implement Object-Oriented Programming principles using ES6 class syntax.
- Capture form inputs (Name, Roll No, Department, CGPA) dynamically.
- Create dynamic HTML DOM elements programmatically using `document.createElement`.
- Render styled profile cards upon button click.

4. THEORY

ES6 (ECMAScript 2015) introduced formal class syntax (`class`, `constructor`, `extends`) to JavaScript.

The Document Object Model (DOM) allows JavaScript to inspect, manipulate, create, and insert HTML elements dynamically at runtime using `document.createElement()`, `appendChild()`, and `innerHTML`.

Application Flow:

User Form Inputs -> ES6 Student Instance -> Dynamic DOM Element Creation -> Mounted Profile Card

5. ALGORITHM / PROCEDURE

1. Design centered form container with input fields for Name, Roll No, Department, and CGPA.
2. Create Generate Profile button and output container element.
3. Define ES6 Student class with constructor setting student properties.
4. Attach onclick event listener to Generate button.
5. Extract input values and instantiate new Student object.
6. Create DOM elements for profile card title and info block.
7. Format student details and append card to output container.

6. PROGRAM

task3.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Student Profile Generator</title>
  <style>
    body {
      font-family: Segoe UI, sans-serif;
      background: #f1f5f9;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
    }

    .form-box {
      background: #ffffff;
      padding: 30px;
      border-radius: 10px;
      box-shadow: 0 4px 12px rgba(0, 0, 0, 0.08);
      width: 350px;
      text-align: center;
    }

    h2 {
      margin: 0 0 15px;
      color: #1e293b;
    }

    input, button {
      display: block;
      width: 100%;
      padding: 10px;
      margin: 10px 0;
      border-radius: 6px;
      border: 1px solid #cbd5e1;
      box-sizing: border-box;
      font-size: 15px;
    }

    button {
      background: #2563eb;
      color: #ffffff;
      border: none;
      font-weight: 600;
      cursor: pointer;
    }

    button:hover {
      background: #1d4ed8;
    }

    .card {
      background: #0f172a;
      color: #f8fafc;
      padding: 20px;
      margin-top: 20px;
      border-radius: 8px;
    }
```

```
    text-align: left;
  }

  .card h3 {
    margin: 0 0 12px;
    color: #38bdf8;
    border-bottom: 1px solid #334155;
    padding-bottom: 6px;
  }

  .profile-info p {
    margin: 8px 0;
    font-size: 15px;
  }
</style>
</head>
<body>

<div class="form-box">
  <h2>Student Details</h2>
  <input type="text" id="name" placeholder="Name">
  <input type="text" id="roll" placeholder="Roll Number">
  <input type="text" id="dept" placeholder="Department">
  <input type="number" step="0.01" id="cgpa" placeholder="CGPA">
  <button id="btn">Generate Profile</button>
  <div id="container"></div>
</div>

<script>
class Student {
  constructor(name, rollNo, department, cgpa) {
    this.name = name;
    this.rollNo = rollNo;
    this.department = department;
    this.cgpa = cgpa;
  }
}

document.getElementById('btn').onclick = function() {
  const s = new Student(
    document.getElementById('name').value,
    document.getElementById('roll').value,
    document.getElementById('dept').value,
    document.getElementById('cgpa').value
  );

  const container = document.getElementById('container');
  container.innerHTML = '';

  const card = document.createElement('div');
  card.className = 'card';

  const title = document.createElement('h3');
  title.innerText = 'Student Profile';

  const infoDiv = document.createElement('div');
  infoDiv.className = 'profile-info';
  infoDiv.innerHTML = `
```

```

    <p><b>Name      :</b> ${s.name}</p>
    <p><b>Roll No   :</b> ${s.rollNo}</p>
    <p><b>Department :</b> ${s.department}</p>
    <p><b>CGPA      :</b> ${s.cgpa}</p>
  `;

  card.appendChild(title);
  card.appendChild(infoDiv);
  container.appendChild(card);
};
</script>

</body>
</html>

```

7. CODE EXPLANATION

Code / Syntax	Explanation
class Student { constructor(...) }	Defines ES6 class template for instantiating student objects.
new Student(name, roll, dept, cgpa)	Instantiates student instance from input field values.
document.getElementById('btn').onclick	Binds click handler to Generate Profile button.
document.createElement('div')	Programmatically creates new div DOM node for card container.
card.appendChild(infoDiv)	Appends child node into profile card container.
container.appendChild(card)	Mounts finished profile card into main page DOM.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Input Form Rendering	Form displays input fields cleanly	Form displays input fields cleanly	Pass
TC-02	Student Class Object Creation	ES6 constructor initializes object attributes correctly	ES6 constructor initializes object attributes correctly	Pass
TC-03	Dynamic Profile Card Generation	Card is created and appended to container on click	Card is created and appended to container on click	Pass
TC-04	Data Value Formatting	Submitted Name, Roll No, Dept, and CGPA appear formatted	Submitted Name, Roll No, Dept, and CGPA appear formatted	Pass
TC-05	Re-generation Clear Action	Previous profile card is replaced when generating new profile	Previous profile card is replaced when generating new profile	Pass

9. OUTPUT

Output 1: Student Input Details Form & Generated Profile Card

The screenshot displays a web application interface. On the left, a 'Student Details' form is visible with the following fields: Name (Sharan), Roll No. (A24120502261), Department (CSE), and CGPA (8.5). Below these fields is a 'Generate Profile' button. To the right of the form, a 'Student Profile' card is shown, displaying the same information in a structured format:

Student Profile	
Name	Sharan
Roll No.	A24120502261
Department	CSE
CGPA	8.5

10. RESULT

Thus, the Student Profile Generator using ES6 JavaScript Classes and dynamic DOM manipulation was successfully created and verified across all test cases.